

E064

Mathematics for Programmers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A cloud job's total cost is logged by a Python script. After 2 hours the cost is \$17; after 6 hours it is \$29. The job runs on a GPU instance in Oregon. At what rate does the cost grow per hour?

- A. About \$4.83 per hour
- B. About \$0.33 per hour
- C. \$3 per hour
- D. \$12 per hour

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A cost model is $y = 3x + 11$, where x is hours. After how many hours will the cost reach 41?

- A. 41
- B. 10
- C. About 13.67
- D. 134

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A script defines $f(x) = 2 * x + 1$ and $g(x) = x ** 2$. It also imports math and sets `DEBUG = True`. What does `print(f(g(3)))` show?

- A. 19
- B. 63
- C. 16
- D. 49

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which pair gives $h(x) = (x + 4)^2$ when $h(x) = f(g(x))$?

- A. $f(x) = x^2$ and $g(x) = 4$, added together
- B. $f(x) = x + 4$ and $g(x) = x + 4$, multiplied together
- C. $g(x) = x^2$ and $f(x) = x + 4$
- D. $g(x) = x + 4$ and $f(x) = x^2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A log dataset starts at 1 GB and doubles in size every month. It is stored in cloud storage managed by a two-person data team. After how many months will it first reach 1,024 GB?

- A. 10 months
- B. 512 months
- C. About 3 months
- D. 11 months

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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6.

A student on a 2019 laptop running Python 3.12 writes `math.sin(30)`, expecting 0.5 for a 30-degree angle, but gets about -0.988 . What is the fix?

- A. Use `math.sin(math.radians(30))`.
- B. Use `math.sin(math.degrees(30))`.
- C. Use `math.sin(30 / 180)`.
- D. Round the result, since it is a floating-point error.

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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7.

A sensor on a factory fan, sampled 1,000 times per second, is modeled by $y = 4\sin(2\pi x) + 1$. What are the amplitude and period of the model?

- A. Amplitude 5, period π
- B. Amplitude 4, period π
- C. Amplitude 4, period 4π
- D. Amplitude 2, period 4

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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8. [TRANSFER — reversed stem / different variable]

You want $y = \sin(B\pi x)$ to repeat every 4π units. What value of B should you use?

- A. 2
- B. 4
- C. 0.5
- D. 4π

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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9.

In a budgeting script a finance intern wrote last spring, the check `0.1 + 0.2 == 0.3` prints False. What is the best explanation and fix?

- A. Convert each number with `int()` first.
- B. Binary floats cannot store 0.1 exactly; compare with `math.isclose`.
- C. It is a Python bug that will be patched.
- D. Never use floats for any calculation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A timer for a fitness app with 12,000 users stores 250 seconds. The code shows `minutes = 250 // 60` and `seconds = 250 % 60`. What does the display read?

- A. About 4.17 minutes
- B. 4 minutes 17 seconds
- C. 10 minutes 4 seconds
- D. 4 minutes 10 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

What does Python print for `print(-7 // 2, -7 % 2)`?

- A. -4 -1
- B. -3 -1
- C. -3.5 1
- D. -4 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A physics game models a ball's height as $h(t) = -5t^2 + 20t + 2$ meters. The launch is shown at 60 frames per second. What is the ball's maximum height?

- A. -58 meters
- B. 2
- C. 22 meters
- D. 20 meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A cloud job's total cost is logged by a Python script. After 2 hours the cost is \$17; after 6 hours it is \$29. ~~The job runs on a GPU instance in Oregon.~~ At what rate does the cost grow per hour?

Target word: rate per hour **Decides:** Slope = $(29 - 17) / (6 - 2)$.

A	Adjacent category	Average cost per hour ($\$29 / 6$), which includes the starting fee, not the rate.
B	Inverted operation	Divides change in x by change in y.
C	Correct	$12 / 4 = 3$.
D	Dropped element	The change in cost, not divided by the 4 hours.

2. Answer B Concept 1 · transfer

A cost model is $y = 3x + 11$, where x is hours. After how many hours will the cost reach 41?

Target word: solve for x **Decides:** $41 = 3x + 11$, so $3x = 30$.

A	Wrong stem number	The target cost, not the hours.
B	Correct	$(41 - 11) / 3 = 10$.
C	Dropped element	$41 / 3$ forgets to subtract the starting value 11.
D	Inverted operation	Plugs 41 in as x instead of solving for x.

3. Answer A Concept 2

A script defines $f(x) = 2 \cdot x + 1$ and $g(x) = x \cdot x$. ~~It also imports math and sets DEBUG = True.~~ What does `print(f(g(3)))` show?

Target word: value of $f(g(3))$ **Decides:** g runs first: $g(3) = 9$, then $f(9)$.

A	Correct	$g(3) = 9$; $f(9) = 2 \cdot 9 + 1 = 19$.
B	Adjacent category	$f(3) \times g(3) = 7 \times 9$ multiplies instead of composing.
C	Confusion set	$f(3) + g(3) = 7 + 9$ adds instead of composing.
D	Inverted operation	That is $g(f(3)) = 7^2$ — outer function applied first.

4. Answer D Concept 2 · transfer

Which pair gives $h(x) = (x + 4)^2$ when $h(x) = f(g(x))$?

Target word: decompose a composition **Decides:** The inner step adds 4; the outer step squares.

A	Confusion set	$x^2 + 4$ is a sum, and it does not equal $(x + 4)^2$.
B	Adjacent category	A product of functions, not a composition.
C	Inverted operation	Gives $x^2 + 4$ — the order is reversed.
D	Correct	$f(g(x)) = (x + 4)^2$.

5. Answer A Concept 3

A log dataset starts at 1 GB and doubles in size every month. ~~It is stored in cloud storage managed by a two-person data team.~~ After how many months will it first reach 1,024 GB?

Target word: number of doublings **Decides:** $2^n = 1,024$, so $n = \log_2(1,024)$.

A	Correct	$2^{10} = 1,024$.
B	Inverted operation	Divides 1,024 by 2 instead of taking a log.
C	Confusion set	$\log_{10}(1,024) \approx 3$ uses the wrong base.
D	Dropped element	Counts the starting month as a doubling.

6. Answer A Concept 4

A student ~~on a 2019 laptop running Python 3.12~~ writes `math.sin(30)`, expecting 0.5 for a 30-degree angle, but gets about -0.988 . What is the fix?

Target word: why the trig result is off **Decides:** `math.sin` takes radians.

A	Correct	Converts 30 degrees to $\pi/6$ radians, giving 0.5.
B	Inverted operation	Converts the wrong way, from radians to degrees.
C	Dropped element	Leaves out π in the conversion.
D	Confusion set	Float error is tiny; this is a unit mistake.

7. Answer B Concept 5

A sensor on a factory fan, ~~sampled 1,000 times per second~~, is modeled by $y = 4\sin(2x) + 1$. What are the amplitude and period of the model?

Target word: amplitude and period **Decides:** $A = 4$, $B = 2$, so period $= 2\pi/2$.

A	Adjacent category	Adds the vertical shift, which moves the graph but does not change amplitude.
B	Correct	$ A = 4$; $2\pi / 2 = \pi$.
C	Inverted operation	Multiplies 2π by B instead of dividing.
D	Confusion set	Swaps the roles of the two numbers.

8. Answer C Concept 5 • transfer

You want $y = \sin(Bx)$ to repeat every 4π units. What value of B should you use?

Target word: solve for B from the period **Decides:** Period $= 2\pi/B$, so $B = 2\pi / 4\pi$.

A	Inverted operation	$4\pi / 2\pi$ flips the ratio.
B	Confusion set	Treats B as the period's coefficient.
C	Correct	$2\pi / 4\pi = 1/2$.
D	Wrong stem number	The period itself, not B.

9. Answer B Concept 6

In a budgeting script ~~a finance intern wrote last spring~~, the check $0.1 + 0.2 == 0.3$ prints False. What is the best explanation and fix?

Target word: cause and fix of failed float equality **Decides:** Binary floats store most decimals approximately.

A	Adjacent category	<code>int()</code> truncates all three values to 0 and loses the data.
B	Correct	Tolerance-based comparison handles tiny representation errors.
C	Confusion set	All IEEE 754 languages behave this way.
D	Absolute language	Floats are fine with a tolerance; exact money math can use Decimal.

10. Answer D Concept 7

A timer ~~for a fitness app with 12,000 users~~ stores 250 seconds. The code shows `minutes = 250 // 60` and `seconds = 250 % 60`. What does the display read?

Target word: floor division and remainder **Decides:** $250 = 4 \times 60 + 10$.

A	Adjacent category	True division (<code>/</code>), not floor division.
B	Confusion set	Reads the decimal part of 4.17 as seconds.
C	Inverted operation	Swaps quotient and remainder.
D	Correct	$250 // 60 = 4$; $250 \% 60 = 10$.

11. Answer D Concept 7 · transfer

What does Python print for `print(-7 // 2, -7 % 2)`?

Target word: floor division with a negative **Decides:** Python floors toward negative infinity.

A	Dropped element	Drops the rule that the remainder takes the divisor's sign.
B	Confusion set	Truncation toward zero, as in C or Java.
C	Adjacent category	True division, not floor division.
D	Correct	$\text{floor}(-3.5) = -4$, and $-4 \times 2 + 1 = -7$.

12. Answer C Concept 8

A physics game models a ball's height as $h(t) = -5t^2 + 20t + 2$ meters. ~~The launch is shown at 60 frames per second.~~ What is the ball's maximum height?

Target word: maximum y value **Decides:** Vertex at $t = -20 / (2 \times -5) = 2$.

A	Inverted operation	Uses $t = b/(2a) = -2$, dropping the minus sign.
B	Adjacent category	The time of the peak, not its height.
C	Correct	$h(2) = -20 + 40 + 2 = 22$.
D	Dropped element	Leaves out the +2 starting height.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Slope as a rate of change	Slope = change in $y \div$ change in x between two points; in $y = mx + b$, m is the rate and b is the starting value.	Dividing one y -value by one x -value, or flipping the ratio.	Two (x, y) readings and “per” in the question.
2	Function composition order	$f(g(x))$ applies g first, then feeds its output into f .	Applying the outer function first.	Nested calls like $f(g(3))$.
3	Logarithms undo exponentials	If $b^y = x$, then $y = \log_b(x)$; use a log to solve for an exponent such as a number of doublings.	Dividing instead of taking a log, or using the wrong base.	Repeated doubling or growth and “how many steps.”
4	Radians in code	Python’s math trig functions take radians; convert degrees with <code>math.radians (degrees $\times \pi/180$)</code> .	Passing degrees straight into <code>math.sin</code> .	A trig call with a familiar degree value giving an odd result.
5	Amplitude and period	For $y = A \cdot \sin(Bx) + D$, amplitude is $ A $, period is $2\pi/ B $, and D only shifts the graph up or down.	Reading B as the period or adding D to the amplitude.	A sine or cosine model with numbers inside and outside.
6	Floating-point precision	Binary floats cannot store most decimals exactly, so compare floats with a tolerance (<code>math.isclose</code>), not <code>==</code> .	Thinking $0.1 + 0.2 == 0.3$ is a Python bug.	An equality check on decimal results that fails.
7	Floor division and modulo	$a // b$ gives the floor of the quotient and $a \% b$ the remainder; in Python the remainder takes the divisor’s sign.	Reading the decimal part of a / b as the remainder, or expecting truncation toward zero.	Converting units (seconds to minutes) or negative operands.
8	Quadratic vertex	For $y = ax^2 + bx + c$, the vertex is at $x = -b/(2a)$; plug that x back in to get the max or min value.	Stopping at the x of the vertex when the question asks for y , or dropping the sign.	“Maximum height” or “minimum cost” with a squared term.

Course scope (public description): This course bridges mathematics and programming by using Python to develop mathematical reasoning skills. Students create and evaluate functions to model relationships, analyze and graph algebraic and trigonometric functions, and solve problems that build readiness for calculus and object-oriented programming while using computational tools to explore patterns, precision, and real-world applications.